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Checklist

What you should know

After studying this subtopic you should be able to:

- Determine rates of reaction from tangents of graphs of concentration, volume or mass against time.
- Explain how changes in reaction rate can be monitored indirectly by monitoring changes in mass, volume or colour.
- Explain the relationship between the kinetic energy of the particles and the temperature in kelvin, and the role of collision geometry.
- Explain the effect of changing conditions on the rate of reaction.
- Sketch Maxwell-Boltzmann energy distribution curves to explain the effect of temperature on the probability of successful collisions.
- Sketch and explain energy profiles with and without catalyst for endothermic and exothermic reactions.
- Construct Maxwell-distribution curves to explain the effect of different values of E_a on the probability of successful collisions.

Higher level (HL)

- Evaluate proposed reaction mechanisms and recognise reaction intermediates.
- Distinguish between intermediates and transition states and recognise both in energy profiles of reactions.
- Construct and interpret energy profiles from kinetic data.
- Interpret the terms unimolecular, bimolecular and termolecular.
- Deduce the rate equation for a reaction from experimental data.
- Identify the number of particles in the rate determining step from the order with respect to a reactant and deduce the overall order of reaction.
- Sketch, identify and analyse graphical representations of zero order, first order and second order reactions.
- Solve problems using the rate equation including finding the units of k.
- Describe the qualitative relationship between temperature and the rate constant.

- Analyse graphical representations of the Arrhenius equation including its linear form.
- Determine the activation energy and Arrhenius factor from experimental data.

Practical skills

Once you have completed this subtopic, go to Practical 6a: Determining the rate of a chemical reaction (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/determining-the-rate-of-a-chemical-reaction-6a-id-46721/) and Practical 6b: Investigating factors affecting the rate of a chemical reaction (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/investigating-factors-affecting-the-rate-of-a-chemical-reaction-6b-id-46722/) to apply your skills to monitor the rate of a chemical reaction using different techniques.